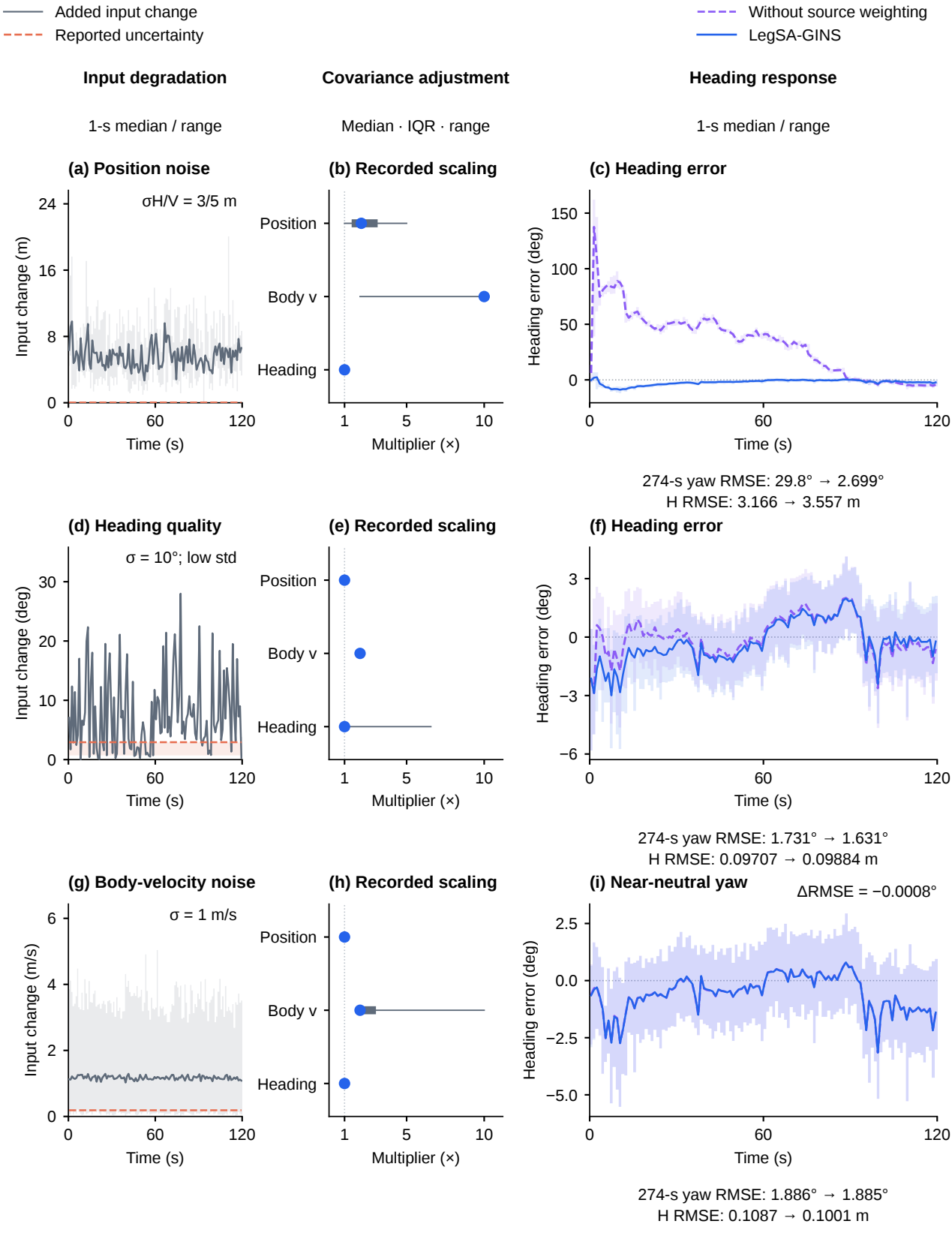

Added input change
Reported uncertainty
Without source weighting
LegSA-GINS
Input degradation
Covariance adjustment
Heading response
1-s median / range
Median · IQR · range
1-s median / range
(a) Position noise
σH/V = 3/5 m
Input change (m)
Time (s)
(b) Recorded scaling
Position
Body v
Heading
Multiplier (×)
(c) Heading error
Heading error (deg)
Time (s)
274-s yaw RMSE: 29.8° → 2.699°
H RMSE: 3.166 → 3.557 m
(d) Heading quality
σ = 10°; low std
Input change (deg)
Time (s)
(e) Recorded scaling
Position
Body v
Heading
Multiplier (×)
(f) Heading error
Heading error (deg)
Time (s)
274-s yaw RMSE: 1.731° → 1.631°
H RMSE: 0.09707 → 0.09884 m
(g) Body-velocity noise
σ = 1 m/s
Input change (m/s)
Time (s)
(h) Recorded scaling
Position
Body v
Heading
Multiplier (×)
(i) Near-neutral yaw
ΔRMSE = −0.0008°
Heading error (deg)
Time (s)
274-s yaw RMSE: 1.886° → 1.885°
H RMSE: 0.1087 → 0.1001 m